

Quote Tool — v1 Companion Guide

A first proof-of-concept of the profiled-waste item-code finder — what it does, what to trust, and what's next.

The problem we're attacking. Today, to quote a profiled (non-lab-pack) waste stream, Kristin scrolls ~760 item codes by hand to find the right one. This tool narrows that to a short list in seconds — ranked by what RedArc has *actually billed* for similar waste, with the cheaper facility options and the reasoning shown. **This is a v1: built to prove the approach is real, not to quote live yet.**

How it works

We joined Zak's five data files into one catalog (761 codes + pricing + routing). The key ingredient is **889 real waste-stream profiles** — each already labeled with the item code RedArc billed and the facility it went to. That's an answer key. So instead of hand-coding every routing rule, the tool:

- Reads your waste description (a manifest shipping line works great — UN number, hazard class, packing group).
- Finds the most similar *real past profiles* and recommends the code *they* were billed under.
- Shows the facility options ranked by cost, plus Zak's routing notes (preferred / backup facility).
- Gives a **confidence flag** — and when it's unsure, it says "confirm with Zak" instead of guessing.

How accurate is it? (measured, not claimed)

We replayed all 889 real profiles through the tool (hiding each one's answer) and checked whether it recommended the code that was actually billed:

72%

right code in the top-3 shortlist

74%

top-1 correct on high-confidence calls

889

real profiles as the answer key

Why top-3 is the number that matters: the job is to narrow 760 codes to a shortlist. Handing Kristin a 3-code shortlist that contains the right answer 72% of the time — from a waste description alone — already replaces most of the scrolling. The confident cases (~40%) can auto-fill; the rest still get a short list instead of the whole catalog. **This is a baseline to beat, and every future improvement is measurable against it.**

Reading the demo

Paste a waste description (or click an example). You'll see the recommended item code, a confidence badge, the facility options (cheapest first, as relative \$ / \$\$ / \$\$\$ bands), Zak's routing notes, and the real waste streams it matched on. Try a manifest shipping line like UN1993, Waste Flammable liquids n.o.s. (acetone), 3, PGII.

Caveats — please read before judging

It's v1 on raw data. Item-code descriptions are inconsistent (generated from vendor rate sheets, never cleaned). We did *not* clean them first — cleanup is a later step. The number above is despite the mess, not because the data is tidy.

The demo is de-identified. No customer/generator names, and prices are shown as relative bands (\$ = cheapest), not real rates — the public dashboard isn't a place for RedArc's pricing. The full-dollar version runs locally on Brian's machine.

The hardest layer isn't in yet. No SDS/chemistry reasoning, cost-optimization, or facility acceptance-criteria (e.g. Tulsa's 500 ppm VOC limit) — and the approvals-department conservatism that lives only in Zak's head is deliberately left as a "low-confidence → confirm" flag, not a guess.

Not for production quoting. This proves direction. It is not yet a tool Kristin should quote a customer from unaided.

A few questions for Zak

1. Does the shape fit? Is "short list of 3 + a confidence flag, auto-fill when sure, confirm-with-Zak when not" the right way for Kristin to work?

2. Two routing corrections to confirm (from your Operational Color notes): the HF→Tulsa code is still in the recycling group (remove?), and recycled petroleum shouldn't be under an FBL code to VAL — master-list error or a rogue profile?

3. Where should it live? A Claude skill Kristin invokes (like the lab-pack sorter), or a small internal web tool? Either is easy.

4. Secured setup? To run on the full real data (live prices, customer-specific rates) it needs a private/secured home. Worth setting up — or keep proving it locally first?

What's next

The accuracy levers we haven't pulled yet, each measurable against the 72%: (1) match on hazard class + RCRA/state codes more deeply, (2) Claude's chemistry/SDS reasoning, (3) the cost-optimization and acceptance-criteria routing layers Zak described. Then: wire in the lab-pack skill so one flow handles both lab packs and profiled waste end-to-end.